A CASE-STUDY BASED REPORT

ON

Travel Time Prediction Using Ridge and Lasso Regression

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by

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1. ABSTRACT

Travel time prediction is a crucial component of intelligent transportation systems, enabling efficient route planning, traffic management, and improved commuter experiences. Accurate prediction of travel time helps reduce traffic congestion, minimizes fuel consumption, and supports informed decision-making for both travelers and transportation authorities. However, predicting travel time is a challenging task due to the influence of multiple dynamic factors such as traffic density, weather conditions, road infrastructure, time of day, and unexpected incidents. This study presents a travel time prediction model using Ridge Regression and Lasso Regression, two widely used regularization techniques that improve the performance of linear regression models by reducing overfitting and handling multicollinearity among input variables.

The proposed approach begins with data preprocessing, including handling missing values, feature selection, normalization, and splitting the dataset into training and testing sets. Ridge Regression applies L2 regularization to shrink coefficient values while retaining all features, making it suitable for datasets with correlated variables. In contrast, Lasso Regression employs L1 regularization, which not only minimizes prediction error but also performs automatic feature selection by assigning zero coefficients to less significant variables. The predictive performance of both models is evaluated using standard regression metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2 score).

The comparative analysis highlights the strengths and limitations of each regression technique in predicting travel time under varying traffic conditions. Results demonstrate that Ridge Regression provides stable predictions in the presence of multicollinearity, while Lasso Regression enhances model interpretability by identifying the most influential features affecting travel time. The findings indicate that both methods offer reliable and computationally efficient solutions for travel time prediction, making them suitable for real-time transportation applications. This research contributes to the development of intelligent traffic prediction systems that can support smarter urban mobility, optimize route planning, and improve the overall efficiency of modern transportation networks.

2. INTRODUCTION

2.1 Background

Transportation is a regular part of daily life for students, employees, business owners, and other travelers. Before starting a journey, people usually want to know how far their destination is, how long the journey may take, and which route they should follow. These requirements become more important when travelling to an unfamiliar place or visiting a particular local business.

Travel time depends on several factors. Distance is an important factor, but it is not the only one. Traffic conditions, weather, departure time, day of the week, and the selected mode of transportation can also affect the journey.

Traditional distance calculations alone cannot fully represent these variations. For example, two journeys of the same distance may take different amounts of time because one occurs during heavy traffic and the other during a less congested period.

Machine learning provides a way to study relationships between travel-related features and historical travel-time data. Regression algorithms can learn patterns from a dataset and estimate travel duration for new inputs. When combined with mapping and routing services, these predictions can be presented through a user-friendly web application.

WayVista applies these concepts by combining a travel-time prediction system with location search, route visualization, and journey planning features.

2.2 Motivation for the Project

The main motivation for developing WayVista is to make travel planning easier and demonstrate a practical application of machine learning.

The project was motivated by the following challenges:

1. Uncertainty in travel time: Users may not know how long a journey will take under different traffic conditions.
2. Difficulty finding local destinations: Users may know a shop or restaurant's name without knowing its exact address.
3. Need for current-location support: Users may prefer to start a journey from their present location instead of entering an address manually.

4. Different modes of transportation: Car, two-wheeler, transit, and walking journeys require different time estimates.
5. Need for route visualization: Users benefit from seeing the source, destination, and route on a map.
6. Practical application of machine learning: The project provides an opportunity to implement and compare Ridge Regression and Lasso Regression on a travel-related prediction problem.

These challenges motivated the development of a system that combines route calculation, travel-time prediction, and an accessible web interface.

2.3 Overview of the Work Carried Out

The project involves the development of a web application and an ML prediction pipeline.

The main work carried out includes:

- Project setup: Organizing the frontend, backend, dataset, and model files.
- Dataset preparation: Inspecting the travel dataset, checking its columns, preparing features, and separating training and testing data.
- Exploratory Data Analysis: Studying distance and travel-time distributions and examining the available features.
- Model development: Training Ridge Regression and Lasso Regression models.
- Model evaluation: Comparing model performance using metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (\(R^2\)).
- Backend development: Using FastAPI to provide application endpoints and process prediction requests.
- Location and routing integration: Using Nominatim for place search, OSRM for routing, and OpenStreetMap with Leaflet for map visualization.
- Weather integration: Using Open-Meteo to obtain weather information.
- Journey planning features: Supporting manual locations, current-location input, destination search, travel-mode estimates, traffic adjustments, and ETA calculation.
- Authentication: Implementing registration, email verification, login, and dashboard access as part of the intended application workflow.

The project also involves testing the route display, input handling, travel-time calculations, and integration between the frontend and backend.

2.4 Scope and Responsible-Use Statement

The scope of WayVista is to demonstrate a practical travel-planning application that combines machine learning with routing and location services.

The system covers:

- Travel-time prediction using Ridge Regression and Lasso Regression.
- Source and destination entry.
- Current-location support, subject to browser permission.
- Place-name searches for destinations such as shops and landmarks.
- Route distance and duration retrieval through the routing service.
- Map visualization and route guidance where supported by the route data.
- Travel-mode estimates and traffic-condition adjustments.
- Estimated arrival-time calculation.
- User registration and email verification.

The system has several limitations. The quality of location search depends on the availability and accuracy of OpenStreetMap data. OSRM route durations do not necessarily match the durations shown by commercial navigation services. Some travel-mode durations in the application are estimates rather than results from dedicated live routing engines. The ML models are limited by the data on which they were trained.

The system should therefore be treated as an educational decision-support tool. Its predictions are estimates, not guarantees. Users should follow applicable traffic rules, verify routes when necessary, and avoid interacting with the application while driving. Current-location access should require user permission and should not involve unnecessary storage of precise location data.

3. PROBLEM STATEMENT

3.1 Case-Study Scenario

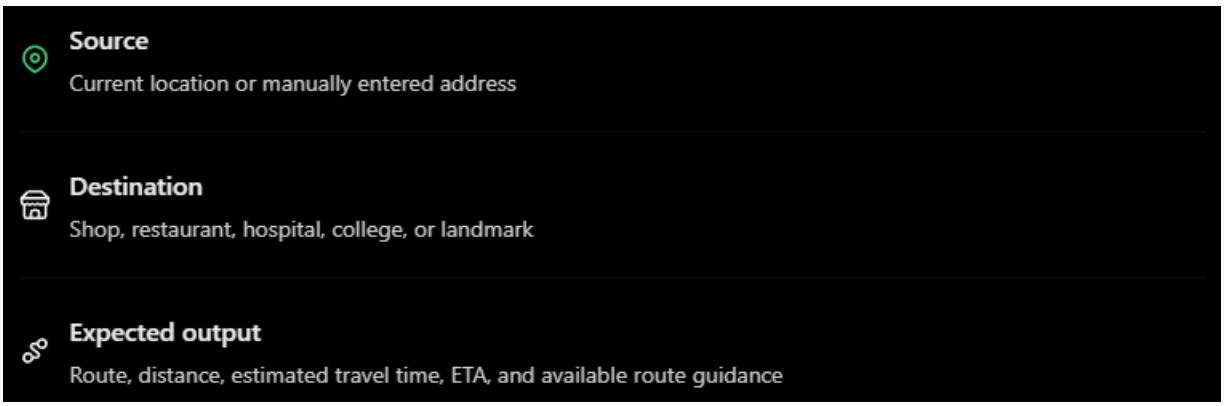
Consider a student living in Hyderabad who wants to visit a shop in an unfamiliar area. The student knows the shop's name but may not know its exact address. Before leaving, the student wants to determine the distance, expected travel time, and arrival time.

The student may also need to compare travel options, such as travelling by car, two-wheeler, public transit, or walking. In addition, traffic conditions and weather may affect the journey.

Without a convenient system, the student may need to search for the destination separately, calculate the route, estimate the duration, and determine the arrival time.

WayVista addresses this scenario through a single web application. The user can choose the current location or enter a source manually, search for a destination by name, view the route, and obtain travel-time estimates.

For example, the journey may be entered as:



This scenario represents the practical problem studied in the project.

3.2 Central Case-Study Question

How can a machine-learning-based travel planning system estimate journey time and provide route guidance for users travelling from their current or manually entered location to a selected destination, while considering distance, traffic conditions, weather, departure time, and mode of transportation?

The study also investigates how Ridge Regression and Lasso Regression can be applied to travel-time prediction and how their results can be integrated into a web application.

3.3 Formal Problem Definition

The project treats travel-time prediction as a supervised regression problem.

Let the input feature vector be represented by:

$$X = (d, t, w, \delta, h, T)$$

where:

- d= distance in kilometres
- t= traffic condition
- w= weather condition
- δ= day of the week
- h= departure hour
- T= temperature in degrees Celsius

The target variable is:

$$y = \text{Travel time in minutes}$$

The objective is to learn a function f that estimates travel time from the available input features:

$$\hat{y} = f(X)$$

Two regression algorithms, Ridge Regression and Lasso Regression, are trained and evaluated for this task.

The routing component solves a related but separate problem: determining a route between the source and destination coordinates and obtaining the associated distance and route duration.

Therefore, the overall system has two distinct outputs:

1. Routing output: Route geometry, distance, and routing-engine duration.
2. ML output: Predicted travel time based on the trained regression model.

These outputs should not be treated as interchangeable because they are produced using different methods.

3.4 Objectives that Guide the Study

The main objectives of the project are:

1. To develop a web-based travel-time prediction system.
2. To prepare and analyse a travel-related dataset.
3. To implement Ridge Regression and Lasso Regression.
4. To evaluate the models using appropriate regression metrics.
5. To allow users to enter source and destination locations.
6. To support destination searches using place and business names where available.
7. To calculate route distance and duration using a routing service.
8. To display the route using an interactive map.
9. To estimate travel time under different traffic conditions.
10. To provide estimated arrival times.
11. To support current-location input, subject to user permission.
12. To integrate the ML prediction system with a FastAPI backend.
13. To provide a simple interface for users to interact with the system.
14. To identify the limitations of the dataset, models, and external services.

3.5 Assumptions and Limitations

Assumptions:

- The dataset contains useful relationships between travel-related features and travel time.
- The selected input features are available when a prediction is requested.
- The user enters a valid source and destination or grants permission to detect their current location.
- The routing and geocoding services are accessible when required.
- The dataset's training and testing subsets are prepared consistently.
- The regression models are used within the limits of their training data.

Limitations:

- The available dataset contains only a limited range of distances.
- Model predictions for routes much longer than the training examples may be unreliable.
- A model trained on a small dataset may not represent every real-world traffic situation.
- Routing-engine durations and ML predictions may differ.
- Place search depends on the coverage of OpenStreetMap data.
- Weather information may not perfectly represent conditions along the entire route.
- Simple traffic multipliers and travel-mode speed estimates are approximations, not live measurements of actual traffic or public-transit services.
- The application requires internet access for its online mapping, geocoding, routing, and weather integrations.

These limitations are important when interpreting the results of the case study**.**

4. Dataset and Exploratory Data Analysis

4.1 About the Dataset

The project uses a CSV dataset named travel_dataset.csv for developing the travel-time prediction models.

The dataset contains 1,000 rows and 7 columns. Six columns are input features, and one column represents the target variable.

Column	Description	Role
distance_km	Travel distance in kilometres	Input
traffic_condition	Traffic category	Input
weather	Weather condition	Input
day_of_week	Day of the week	Input
departure_hour	Hour of departure	Input
temperature_c	Temperature in Celsius	Input
travel_time_minutes	Travel duration in minutes	Target

The target variable is travel_time_minutes, which the models learn to predict from the input features.

The dataset is used for model training and evaluation. External routing data obtained during live use serves a different purpose: finding routes and calculating route-based estimates.

4.2 Purpose of EDA

Exploratory Data Analysis (EDA) is performed to understand the dataset before training the machine-learning models.

The main purposes of EDA are:

- To understand the dataset's size and structure.

- To identify the input and target variables.
- To inspect the distribution of travel distances.
- To study the distribution of travel times.
- To identify missing or invalid values.
- To examine numerical and categorical features.
- To identify possible outliers and unusual observations.
- To understand whether the available data is suitable for regression.
- To prepare the dataset for model training and evaluation.

EDA helps identify the limitations of the available data before conclusions are drawn from model performance.

4.3 Preparation Steps Actually Performed

The following preparation steps describe the dataset and model pipeline documented for this project.

Step 1: Load the dataset

The CSV file is loaded using Python data-processing tools.

Step 2: Inspect the dataset

The number of rows and columns, column names, data types, and sample records are examined.

Step 3: Identify the features and target

The six input features are separated from the target column, travel_time_minutes.

Step 4: Inspect data quality

The dataset is checked for missing values, duplicate records, and invalid numerical values. Any cleaning or removal of records should be reported only if it was actually performed in the code.

Step 5: Prepare categorical features

Categorical variables such as traffic condition, weather, and day of the week are converted into numerical representations suitable for regression.

Step 6: Prepare numerical features

Numerical features are processed using the project's preprocessing pipeline. Scaling is used where appropriate for the regularized regression algorithms.

Step 7: Split the dataset

The dataset is divided into training and testing subsets.

The documented split is:

Subset	Records	Percentage
Training data	800	80%
Testing data	200	20%
Total	1,000	100%

The training data is used to fit the models, while the testing data is used to evaluate their performance on unseen records.

Step 8: Prepare the model inputs

The preprocessing pipeline converts the features into the format expected by Ridge Regression and Lasso Regression.

4.4 EDA Exploration – Implementation

EDA is conducted using Python libraries such as Pandas, NumPy, and Matplotlib, as applicable to the implemented analysis.

The main exploratory analyses include the following.

A. Dataset structure

The dataset is inspected to identify its dimensions, features, data types, and target variable.

B. Descriptive statistics

Measures such as minimum, maximum, mean, median, and standard deviation are calculated for numerical columns.

C. Distance distribution

The distance_km column is analysed to understand the range of journey distances represented in the dataset.

D. Travel-time distribution

The travel_time_minutes column is analysed to understand typical journey durations and their variation.

E. Relationship between distance and travel time

Distance and travel time can be plotted using a scatter plot to examine whether longer journeys generally correspond to longer travel durations.

F. Categorical feature inspection

The available categories for traffic condition, weather, and day of the week are examined to understand the types of observations represented in the data.

G. Feature preparation

Numerical and categorical variables are processed for use by the regression algorithms.

These analyses help establish whether the data has sufficient variation and whether additional data may be needed for more reliable predictions.

4.5 Summary of EDA Findings

The dataset statistics documented for this project are summarised below.

Dataset size **1,000** Records	Total columns 7 6 inputs + 1 target
Mean distance 26.34 km	Mean travel time 70.07 min

Statistical measure	Distance	Travel time
Minimum	1.02 km	3.00 min
Maximum	49.98 km	234.64 min
Mean	26.34 km	70.07 min
Median	26.98 km	65.87 min
Standard deviation	13.96 km	44.14 min

The following conclusions can be drawn from these statistics:

1. The dataset covers travel distances of approximately 1–50 km.
2. The mean travel time is approximately 70 minutes.
3. The standard deviation of travel time indicates considerable variation in journey duration.
4. The dataset provides a basis for learning the relationship between the selected features and travel time.
5. The limited distance range is an important constraint when applying the trained model to long-distance journeys.

Important finding: The dataset's maximum distance is approximately 50 km. Therefore, predicting travel times for journeys of several hundred kilometres is an extrapolation beyond the training range. Such predictions should not be interpreted as reliably validated long-distance estimates.

The EDA findings support the use of regression models for the educational prediction task, while also identifying the need for a larger and more diverse dataset for broader real-world use.

5. Algorithm Selection and Implementation

5.1 Selection of Algorithms

Travel-time prediction is a regression problem because the output is a continuous numerical value measured in minutes.

Ridge Regression and Lasso Regression were selected because both extend linear regression with regularization. Regularization helps control model complexity and can reduce sensitivity to the training data.

The two algorithms also provide an opportunity to compare predictive performance using the same dataset, features, and evaluation procedure.

5.2 Ridge Regression

Ridge Regression is a linear regression algorithm that applies L2 regularization. It adds a penalty based on the squared magnitude of the model coefficients.

Its objective function can be represented as:

$$\min_{\beta}\left[\sum_{i=1}^{n}(y_i-\hat{y}_i)^2+\alpha\sum_{j=1}^{p}\beta_j^2\right]$$

where:

- $y_i$represents the actual travel time.
- $\hat{y}_i$represents the predicted travel time.
- $\beta_j$represents a model coefficient.
- αcontrols the strength of regularization.
- $n$is the number of observations.
- $p$is the number of input features after preprocessing.

Ridge Regression penalizes large coefficients and generally shrinks them towards zero without forcing them to become exactly zero.

Reason for selection: It provides a regularized baseline for predicting travel time from several potentially related input features.

5.3 Lasso Regression

Lasso Regression is a linear regression algorithm that applies L1 regularization. It adds a penalty based on the absolute values of the model coefficients.

Its objective function can be represented as:

$$\min_{\beta} \left[\sum_{i=1}^{n} (y_i - \hat{y}_i)^2 + \alpha \sum_{j=1}^{p} | \beta_j | \right]$$

Lasso Regression can shrink some coefficients to exactly zero. This property can help with feature selection, although it does not guarantee that any coefficient will become zero in every trained model.

Reason for selection: It allows the project to compare an L1-regularized regression model against Ridge Regression.

5.4 Implementation Methodology

The model implementation follows these steps:

1. Load the travel dataset.
2. Separate the input features and target variable.
3. Split the data into training and testing subsets.
4. Preprocess numerical and categorical features.
5. Fit the preprocessing pipeline using the training data.
6. Transform the training and testing data consistently.
7. Train the Ridge Regression model.
8. Train the Lasso Regression model.
9. Select regularization parameters using cross-validation.
10. Generate predictions for the testing data.
11. Calculate evaluation metrics.
12. Save the trained models and preprocessing objects.
13. Load the saved objects in the FastAPI prediction service.

The preprocessing pipeline is important because the live application must transform new inputs in the same way as the training data.

5.5 Model Evaluation Metrics

The models are evaluated using MAE, RMSE, and R^2.

Mean Absolute Error (MAE) measures the average absolute difference between actual and predicted travel times.

$$MAE = \frac{1}{n}\sum_{i=1}^{n} | y_i - \hat{y}_i |$$

A lower MAE indicates smaller average prediction errors.

Root Mean Squared Error (RMSE) measures the square root of the mean squared prediction error.

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}$$

RMSE penalizes larger errors more strongly than MAE.

Coefficient of Determination (R^2) measures how well the model explains variation in the target variable relative to a mean-based baseline.

$$R^2 = 1 - \frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{n} (y_i - \bar{y})^2}$$

An R^2value closer to 1 indicates a stronger fit on the evaluated dataset, but it does not guarantee that predictions will be accurate for every real-world journey.

5.6 Experimental Results

The documented results from the current model artifacts are as follows.

Evaluation metric	Ridge Regression	Lasso Regression
Optimal alpha	2.1544	0.0187
Training R^2	0.9111	0.9111
Testing R^2	0.9104	0.9105
Testing MAE	9.6254 min	9.6355 min
Testing RMSE	12.8281 min	12.8201 min

The testing R^2values correspond to approximately 91.04% for Ridge Regression and 91.05% for Lasso Regression. These percentages describe the coefficient of determination; they should not be described as guaranteed prediction accuracy.

The results indicate that the two models have very similar performance on the testing subset.

- Ridge Regression has a slightly lower MAE.
- Lasso Regression has a slightly lower RMSE and a marginally higher R^2.
- Neither model is clearly superior across all three metrics.

The small difference suggests that both algorithms perform similarly on this dataset. Further testing with a larger and more diverse dataset would be needed before drawing conclusions about performance on other locations or traffic conditions.

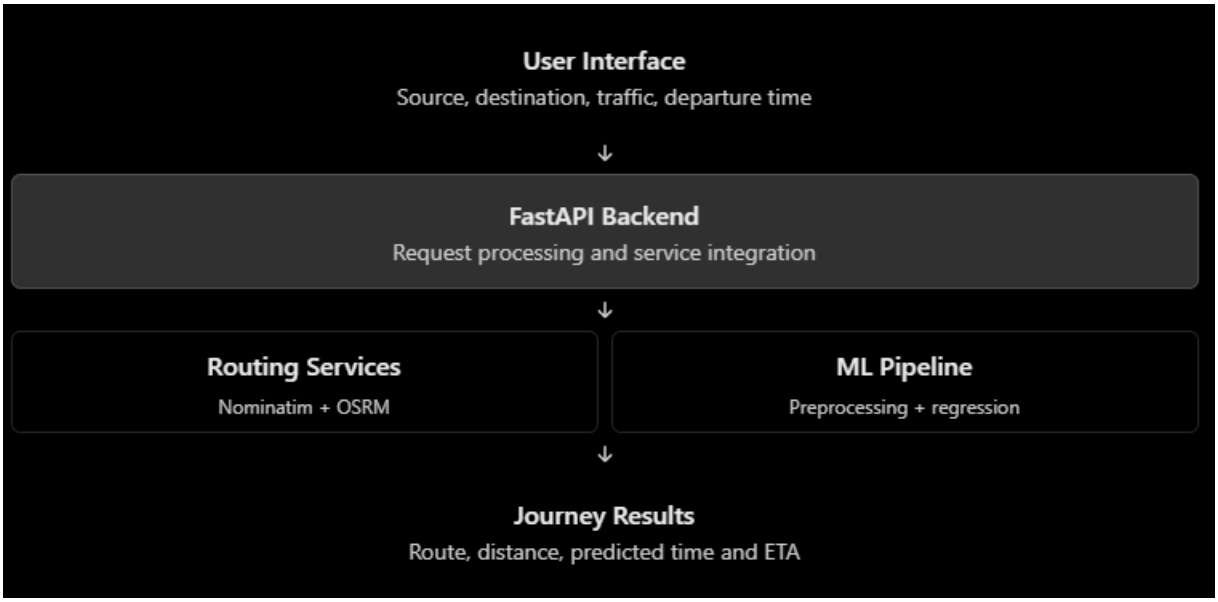
5.7 Integration with the Web Application

The trained models are integrated into the WayVista application through the FastAPI backend.

The frontend collects the journey inputs and sends them to the backend. The backend processes the request, obtains route information where required, prepares the model features, and generates predictions using the saved model and preprocessing objects.

The output is returned to the frontend for display.

The overall flow is:



The map is displayed using Leaflet and OpenStreetMap. Weather information is obtained from Open-Meteo. The system keeps routing-based durations and model-generated predictions distinct.

5.8 Limitations of the Algorithm Implementation

Although the testing results are encouraging, several limitations must be considered.

First, the dataset contains only 1,000 records and covers distances up to approximately 50 km. It may not represent the full variety of traffic and travel conditions encountered in practice.

Second, the regression models learn from the patterns present in the dataset. They do not automatically know the actual traffic conditions of a newly entered route unless appropriate data is supplied.

Third, the routing engine and the ML models solve different tasks. OSRM supplies route information and a routing duration, whereas the regression models estimate travel time from learned feature relationships.

Finally, a high testing R^2 value does not guarantee the same performance on longer journeys, new cities, or data with different characteristics. The results should therefore be interpreted within the scope of the available dataset.

6. Conclusion

WayVista demonstrates the development of a smart travel-time prediction and route-guidance application using machine learning, web development, and location-based services. The project addresses the practical challenge of estimating journey duration and finding destinations when users may not know the exact address.

The system combines Ridge Regression and Lasso Regression with a FastAPI backend and an interactive web interface. The ML component learns relationships between travel-related features and travel time, while Nominatim, OSRM, Leaflet, and OpenStreetMap support destination search, route calculation, and map visualization. Open-Meteo provides weather information.

The dataset contains 1,000 records and seven columns, including six input features and one target variable. The documented testing results show that Ridge Regression and Lasso Regression perform similarly, with testing R^2 values of 0.9104 and 0.9105, respectively. Ridge Regression has a slightly lower MAE, while Lasso Regression has a slightly lower RMSE. These results demonstrate that both algorithms can model the patterns represented in the available dataset.

The project also illustrates the importance of integrating ML predictions with practical software features such as location search, traffic-based estimates, travel modes, ETA calculation, authentication, and route guidance. However, the limited dataset size and distance range mean that further data collection and testing are required before the models can be relied upon for broader real-world travel predictions.

In conclusion, WayVista provides an educational demonstration of how machine learning and routing technologies can work together to support travel planning. Future improvements could include collecting a larger real-world dataset, evaluating models across different regions, improving mode-specific routing, and incorporating better traffic observations.

Report note: The dataset statistics and model metrics above reflect the previously documented project artifacts. Before final submission, verify that the EDA plots, preprocessing steps, email verification, and navigation features described in the report match the final code and have been tested successfully.